

MUSTAFA BAYIRLI

613-890-7611 | mustafabayirli88@gmail.com | [linkedIn.com/in/mustafa](https://www.linkedin.com/in/mustafa) | [Portfolio](#) | Ottawa, ON

Systems-Driven Computer Engineer with hands-on experience in RTL design, digital verification, and hardware/software integration. Passionate about applying expertise in SystemVerilog, testbench development, and functional verification to advance cutting-edge SoC/IP designs for automotive, IoT, and networking applications.

EDUCATION

Bachelor of Applied Science: Computer Engineering with Co-op Program

University of Ottawa

Ottawa, ON

TECHNICAL SKILLS & QUALIFICATIONS

Hardware Design & Verification:

- FPGA/ASIC development (Quartus II, Vivado, EDA), RTL design, VHDL, Verilog, SystemVerilog
- Testbench development, Functional verification, Constrained random testing, RTL code coverage analysis, Simulation debugging (ModelSim), Test plan development
- Circuit Analysis: Analog/digital circuit design, CMOS/RS-232 interfaces, Signal integrity validation
- Basic knowledge of L2/L3 protocols (Ethernet/IP/UDP), UART/SPI protocols

Software Development & Tools: Python, C/C++, Java, Unix/Linux, Git, Bitbucket, Trello, Doxygen

PCB Design: Altium Designer (Certified, 2024), Cadence Allegro, Siemens Xpedition/PADS, high-speed design, ODB++ file generation, Design for manufacturability (DFM)

Testing & Troubleshooting: Oscilloscopes, DMMs, waveform generators, hardware test plans, Lab validation, Logic analyzers

WORK EXPERIENCES

Junior IT Analyst (Co-op)

Innovation Science and Economic Development Canada

Sept 2022 – Dec 2022

Ottawa, ON

- Optimized IT systems by analyzing metrics, reducing troubleshooting time by 15%.
- Authored technical documentation, improving team communication and efficiency by 20%.

Analytical Lab Management (Co-op)

National Research Council Canada (NRC)

Jan 2022 – April 2022

Ottawa, ON

- Designed lab workflows, improving testing efficiency by 25% through systematic analysis.
- Trained teams on hardware/software integration, increasing compliance by 30%

Software Developer (Co-op)

University of Ottawa

July 2021 – Oct 2021

Ottawa, ON

- Converted legal contracts into Symboleo specifications, integrating automated compliance checks.
- Assisted software developers in testing and validating Symboleo specifications.
- Developed sets of LTL/CTL properties to check data protection clauses from a privacy regulation on nuXmv model.

ACADEMIC PROJECTS

FPGA UART | *Digital Systems Course Project*

- Designed a traffic light controller using VHDL and Finite State Machine (FSM), reducing signal errors by 20%.
- Implemented UART protocol on an Altera DE2-115 FPGA, enabling serial data transmission via CMOS-to-RS-232 conversion.
- Validated system integrity through lab testing, ensuring reliable signal processing.

Emergency Urban Search and Rescue Robot | *Capstone Project*

- Led hardware/software integration for an autonomous robot with custom PCBs, Raspberry Pi4, and AWS cloud.
- Developed a React/Node.js web interface for real-time control, demonstrating end-to-end system design.

IDP Simulator & Data Collection | *Real Time System Course Project*

- Developed a real-time solar nano-grid simulator using BeagleBone Black, managing GPIO/ADC/SPI interfaces in C.
- Documented system architecture with Doxygen and tracked tasks via Agile tools (Bitbucket, Trello).